Supplementary Materials: Multi-Mechanism Pre-Training of Physics-Informed Neural Networks for Fouling Resistance Reconstruction from Sparse Measurements

July 8, 2026

Table S1: Auto ODE Selection Ablation — Per-Trial Results

Auto-selection uses a geometric heuristic operating on $N = 15$ sparse points only: ratio of mean R_f in the first third vs. last third of sparse observations. Ratio < 0.15 selects `induction_linear`; otherwise `ks_simple`. Overall accuracy: 87.0% (47 correct / 54 trials across 18 curves × 3 seeds).

Note: On 3 curves (source_003_fig13_pat3, source_003_run2, source_003_run3), the auto-selected "wrong" ODE (ks_simple) achieves *higher* R^2 than the correct ODE in some seeds—the simpler asymptotic prior provides stronger regularization when the data are both sparse and noisy.

Table S2: Spline and Gaussian Process Baselines

All methods use only the N sparse points for fitting and are evaluated on the full curve. Results aggregated across 18 curves × 3 seeds = 54 trials per N value. Median and IQR are computed from all 54 trials.

Note: Cubic spline produces catastrophic oscillations on ∼10–15% of trials (mean is pulled to large negative values, ~ -3000 at $N = 15$), making the mean misleading. The median better reflects typical performance. Smoothing spline and GP variants are more stable. GP-Matérn with $\nu = 2.5$ achieves the highest median R^2 at all N levels.

Comparison with PINN: At $N = 15$, PINN (Main: PT+Phys, single seed 42) achieves median R^2 across curves of approximately 0.94–0.96. Smoothing spline and GP-Matérn achieve median $R^2 \approx 0.99$. However, these methods provide no physics constraints, cannot be used for cleaning optimization, and can produce oscillatory or non-monotonic reconstructions that violate known fouling physics.

Table 1: Auto ODE selection results ($N = 15$, seeds 200–202). Bold rows indicate $|\Delta R^2| > 0.02$.

Curve	Seed	Correct ODE	Auto-selected ODE	Match?	R^2 (Correct)	R^2 (Auto)	Δ
source_001_fig7	200	ks_simple	ks_simple	Y	0.9304	0.9304	0.00
source_001_fig7	201	ks_simple	ks_simple	Y	0.8928	0.8901	0.00
source_001_fig7	202	ks_simple	ks_simple	Y	0.8911	0.8911	0.00
source_002_cba	200	ks_simple	ks_simple	Y	0.3551	0.3551	0.00
source_002_cba	201	ks_simple	ks_simple	Y	0.3053	0.3053	0.00
source_002_cba	202	ks_simple	ks_simple	Y	0.3285	0.3285	0.00
source_002_fed	200	ks_simple	ks_simple	Y	0.6316	0.6316	0.00
source_002_fed	201	ks_simple	ks_simple	Y	0.6231	0.6231	−0.00
source_002_fed	202	ks_simple	ks_simple	Y	0.6587	0.6587	0.00
source_003_coating_l1	200	induction_linear	induction_linear	Y	0.9858	0.9858	0.00
source_003_coating_l1	201	induction_linear	induction_linear	Y	0.9900	0.9900	0.00
source_003_coating_l1	202	induction_linear	induction_linear	Y	0.9887	0.9887	0.00
source_003_coating_l2	200	induction_linear	induction_linear	Y	0.9778	0.9778	0.00
source_003_coating_l2	201	induction_linear	induction_linear	Y	0.9843	0.9843	0.00
source_003_coating_l2	202	induction_linear	induction_linear	Y	0.9829	0.9826	0.00
source_003_coating_l3	200	induction_linear	induction_linear	Y	0.9940	0.9940	0.00
source_003_coating_l3	201	induction_linear	induction_linear	Y	0.9940	0.9942	−0.00
source_003_coating_l3	202	induction_linear	induction_linear	Y	0.9940	0.9940	0.00
source_003_coating_ref	200	induction_linear	induction_linear	Y	0.9970	0.9970	0.00
source_003_coating_ref	201	induction_linear	induction_linear	Y	0.9966	0.9966	0.00
source_003_coating_ref	202	induction_linear	induction_linear	Y	0.9970	0.9970	0.00
source_003_fig11_diapro	200	induction_linear	induction_linear	Y	0.9860	0.9856	0.00
source_003_fig11_diapro	201	induction_linear	induction_linear	Y	0.9882	0.9878	0.00
source_003_fig11_diapro	202	induction_linear	induction_linear	Y	0.9880	0.9882	−0.00
source_003_fig11_grit220	200	induction_linear	induction_linear	Y	0.9903	0.9911	−0.00
source_003_fig11_grit220	201	induction_linear	induction_linear	Y	0.9926	0.9908	0.00
source_003_fig11_grit220	202	induction_linear	induction_linear	Y	0.9904	0.9889	0.00
source_003_fig11_grit80	200	induction_linear	induction_linear	Y	0.9862	0.9880	−0.00
source_003_fig11_grit80	201	induction_linear	induction_linear	Y	0.9920	0.9866	0.00
source_003_fig11_grit80	202	induction_linear	induction_linear	Y	0.9915	0.9904	0.00
source_003_fig11_ref	200	induction_linear	induction_linear	Y	0.9950	0.9990	−0.00
source_003_fig11_ref	201	induction_linear	induction_linear	Y	0.9953	0.9953	−0.00
source_003_fig11_ref	202	induction_linear	induction_linear	Y	0.9899	0.9903	−0.00
source_003_fig13_flat	200	induction_linear	induction_linear	Y	0.9920	0.9920	0.00
source_003_fig13_flat	201	induction_linear	induction_linear	Y	0.9920	0.9921	−0.00
source_003_fig13_flat	202	induction_linear	induction_linear	Y	0.9925	0.9925	0.00
source_003_fig13_pat1	200	induction_linear	induction_linear	Y	0.9661	0.9661	0.00
source_003_fig13_pat1	201	induction_linear	induction_linear	Y	0.9657	0.9656	0.00
source_003_fig13_pat1	202	induction_linear	induction_linear	Y	0.9654	0.9654	0.00
source_003_fig13_pat2	200	induction_linear	ks_simple	N	0.9732	0.9386	**0.03**
source_003_fig13_pat2	201	induction_linear	induction_linear	Y	0.9467	0.9477	−0.00
source_003_fig13_pat2	202	induction_linear	induction_linear	Y	0.9731	0.9731	0.00
source_003_fig13_pat3	200	induction_linear	ks_simple	N	0.7712	0.8579	−0.08
source_003_fig13_pat3	201	induction_linear	ks_simple	N	0.5069	0.9846	−0.47
source_003_fig13_pat3	202	induction_linear	ks_simple	N	0.3710	0.4582	−0.08
source_003_run1	200	induction_linear	induction_linear	Y	0.8532	0.8295	**0.02**
source_003_run1	201	induction_linear	induction_linear	Y	0.8304	0.8304	0.00
source_003_run1	202	induction_linear	induction_linear	Y	0.8371	0.8436	−0.00
source_003_run2	200	induction_linear	ks_simple	N	0.8461	0.8828	−0.03
source_003_run2	201	induction_linear	ks_simple	N	0.8220	0.9442	−0.12
source_003_run2	202	induction_linear	induction_linear	Y	0.6104	0.6104	0.00
source_003_run3	200	induction_linear	induction_linear	Y	0.8563	0.8563	0.00
source_003_run3	201	induction_linear	ks_simple	N	0.9104	1.0300	−0.11
source_003_run3	202	induction_linear	induction_linear	Y	0.8094	0.8094	0.00

Table 2: Spline and GP baseline performance at each sparse budget ($N = 5, 10, 15, 20, 30$). Values are all-trial median [IQR].

Method	$N = 5$	$N = 10$	$N = 15$	$N = 20$	$N = 30$
Cubic Spline	0.52 [0.36, 0.88]	0.94 [0.74, 0.99]	0.99 [0.92, 1.00]	0.99 [0.97, 1.00]	1.00 [0.99, 1.00]
Smoothing Spline	0.84 [0.60, 0.95]	0.98 [0.89, 0.99]	0.99 [0.98, 1.00]	0.99 [0.98, 1.00]	0.99 [0.99, 1.00]
GP-RBF	0.86 [0.66, 0.96]	0.98 [0.90, 0.99]	0.99 [0.98, 1.00]	1.00 [0.99, 1.00]	1.00 [0.99, 1.00]
GP-Matérn	0.89 [0.70, 0.97]	0.98 [0.92, 0.99]	0.99 [0.98, 1.00]	1.00 [0.99, 1.00]	1.00 [1.00, 1.00]